NAMAN TYAGI

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Backend Software Engineer | AI/GenAI Applications | NYU M.S. Computer Science

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL
Backend: FastAPI, Django REST, Redis, PostgreSQL, Celery, REST APIs, Concurrency, Caching
AI/GenAI: LangChain, FAISS, RAG, Transformers, PyTorch, Embeddings, Semantic Search, OCR
Cloud & Data: AWS, Azure, Kafka, Spark Streaming, S3, Blob Storage, SQLite, MongoDB
DevOps & Tools: Docker, GitHub Actions, pytest, Prometheus, Grafana, OpenTelemetry, Git, Ollama, Cursor, Copilot

EXPERIENCE

Backend Developer Intern Sep 2026 – Present
GBCS Group LTD / SkyIT *Remote*

- Developing backend features using Django REST Framework/Python and Express/JavaScript, building and maintaining REST and GraphQL APIs.
- Working with PostgreSQL, MySQL, and MongoDB for schema design, query optimization, and API integration.
- Contributing to Azure deployments, Dockerized services, CI pipelines, code reviews, and backend performance tuning.

Software Engineer Jan 2026 – Aug 2026
NYU Office of Faculty Affairs *New York, NY*

- Built Django REST APIs and PostgreSQL workflows for 500+ faculty, improving system reliability.
- Automated administrative workflows using Celery Beat with escalation logic, reducing manual turnaround time.
- Designed an Excel ingestion pipeline with zero data errors across 20+ test cases; integration tests surfaced 3 pre-release defects.
- Reduced document lookup time by ~60% using AI classification across AWS S3 and Azure Blob Storage.
- Developed a LangChain/FAISS RAG chatbot enabling faculty policy search with semantic retrieval.

AI and Power BI Intern Jun 2025 – Aug 2025
Mast-Jägermeister US, Inc. *White Plains, NY*

- Deployed an Azure-based LangChain/FAISS RAG chatbot for 50+ staff to search 1,000+ documents, reducing retrieval time by 70%.
- Automated ingestion, embedding generation, and semantic indexing for continuous knowledge base updates.
- Improved semantic search accuracy by tuning chunking strategy and adjusting embedding parameters across document types.

Artificial Intelligence Researcher (Part-time) Sep 2023 – Aug 2024
University of Essex *UK (Remote)*

- Fine-tuned transformer NLP models on ~10,000 annotated samples, evaluating precision-recall tradeoffs for publication.
- Published research in Human Behavior and Emerging Technologies and presented at IEEE venues.

Global Technical Support Engineer Jan 2024 – Jun 2024
Ciena *Gurugram, India*

- Resolved 50+ monthly network escalations for 15+ global clients using packet captures, logs, and CLI diagnostics.
- Authored automation scripts and technical documentation that reduced case resolution time by 25%.

PROJECTS

ConTicx | *Next.js, React, TypeScript, Upstash Redis, Hyperswitch, Vercel* | GitHub Aug 2026

- Built a ticketing platform with 10-min Redis inventory holds, server-side pricing, refunds, and payment reconciliation.
- Implemented atomic Redis reservations and concurrency tests (15 requests vs. 10 tickets).

Immune Cell Population Analysis | *Python, SQLite, pandas, Streamlit, pytest, GitHub Actions* | GitHub Aug 2026

- Built a reproducible data-analysis pipeline and normalized SQLite database for clinical-trial immune-cell data.
- Automated data validation, and a 15-test pytest suite with CI integration.

Prism | *Python, FastAPI, PostgreSQL, Redis, Docker Compose, Prometheus, Grafana, Open Telemetry* | GitHub Jul 2026

- Built a production-grade LLM gateway with Redis caching, request-cost tracking, and circuit breaker failover.
- Achieved 245–259 req/sec in Locust load tests; resolved Redis connection bottleneck near 300 req/sec via shared pooling.

RAGBase | *Python, LangChain, FAISS, TF-IDF, Hugging Face, Ollama, Streamlit, Tesseract OCR* | GitHub Apr 2026

- Built a fully offline RAG system supporting PDFs, spreadsheets, images, and scanned documents.
- Added cross-encoder reranking, OCR, and benchmarking on a 1,200-document corpus.

RESEARCH & EXTRACURRICULARS

Computer Vision Research | *Flexible AI-Enabled Mechatronic Systems (FAMS) Lab, NYU Tandon* Fall 2025 – Present

- Developing a real-time computer vision pipeline for detecting kidney stones from continuous microscopic video.
- Preparing image/video datasets, developing and evaluating computer vision and ML approaches, and integrating live inference.

EDUCATION

NYU Tandon School of Engineering - M.S. Computer Science Aug 2024 - May 2026

- Coursework: Algorithms, Machine Learning, Databases, Big Data, Cloud Computing

Amity University (Noida, India) - B.Tech. Computer Science & Engineering (Honors in AI & ML) Jul 2020 – Jun 2024

- Publications: 1 journal article, 1 IEEE conference paper, 2 book chapters